# Sanaul Haque

## Saadya_Polavarapu_Proposal

Course 201

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**Research Topic: Impact of Sleep Disturbance on Gut Brain Axis Among 1st Professional Year MBBS Students in A Tertiary Care Medical College**

**Title: Impact of Sleep Disturbance on Gut Brain Axis Among 1st Professional Year MBBS Students in A Tertiary Care Medical College**

**Existing literature:**

Sleep is a foundation of health for a mental and physical well-being. It is crucial in memory building as reported by Hassan et al [1]. The connection between microbiota and Gut brain axis (GBA) enables communication from gut microbiome to the brain and back making it a bidirectional relationship through neurotransmitter regulation, neuroimmune interaction, neuroendocrine control and microbial metabolite signaling shaping the body's circadian rhythms affecting cognitive functions. [2] – [3].

With development of microbiome research over the last decade, it has demonstrated microbiome produce various neurotransmitters like serotonin, dopamine and gamma – aminobutyric acid (GABA) which affect the mood, anxiety and emotional well-being [1] – [4]. The gut microbiota communicates with sleep related nuclei cells and activates vagus nerve and release neurotransmitters and hormones. [5]

Sleep deprivation decreases gut microbiota and downregulates mucin production and causes altered intestinal barrier function. This provokes systemic inflammation. [2], [6].

In early childhood gut microbiome shows low diversity with Firmicutes to Bacteroides promoting healthy gut environment. Diversity increases with solid food intake with increasing production of short chain fatty acids (SCFA) enhancing gut integrity and immunity. [5]

Undergraduate students experience lifestyle changes with respect to irregular eating, high caffeine intake, inadequate sleep, stress of academic workload. These negatively affect the gut microbiota and lead to gastrointestinal disorders such as bloating, gastritis, constipation. [1], [6]

When beneficial flora dominates, there is production of SCFA regulating melatonin production and circadian rhythm. [5] Dysbiosis can exacerbate the inflammatory process by microbial translocation. [2] - [6] Reduced SCFA levels leas to stress hormone production enhancing sleep deprivation. The Swiss National Science Foundation has reported positive correlation with irregular sleep patterns and gut microbiota. [5]

**Research Problem**:

Scientific evidence links sleep health to microbial composition and function where short chain fatty acids production is a mediator of sleep brain interaction [7]. Existing literature addresses the effect of poor gut health on sleep and academic performances of the young

minds rather than impact of sleep deprivation on gut health. There is a lack of research that integrates sleep deprivation with lifestyle factors unique to undergraduate student population.

Without understanding the irregular sleep schedules, dietary habits and how they alter gut microbiome, wellness interventions in the campus can be ineffective. Therefore, this study assesses the need to understand direct impact of sleep quality on gut brain axis.

**Aim:** To study the impact of sleep disturbances on gut brain axis among 1st Professional Year MBBS students in a tertiary care medical college

**Objectives:**

- To study the sleep pattern among 1st professional year MBBS students using Pittsburgh Sleep Quality Index Questionnaire (PSQI)
- To study Gut health among 1st professional year MBBS students using Rome IV Diagnostic Questionnaire
- To study the diet and lifestyle factors among 1st professional year MBBS students
- To study the relationship between sleep and gut health

**Materials and Methods:**

- **Study design:** Cross sectional study
- **Study Population:** 1st Professional Year MBBS Students
- **Study Duration:** 2 months
- **Study Setting:** Katuri Medical College & Hospital
- **Study size:** 150 students
- **Ethical Committee Approval:** Institutional Ethical Committee approval will be taken prior to the study
- **Inclusion Criteria:** Students with written informed consent
- **Exclusion Criteria:**
  - Students not involved in the sessions
  - Students not willing to participate
  - Students taking medications to influence sleep patterns
- **Data Collection:** Data Survey will be collected using a pre validated structured questionnaire by google forms after explaining that confidentiality. Anonymity of the participant will be maintained.

- Section A will collect sociodemographic data and academic profile.
- Section B will collect data on sleep quality and disturbances from PSQI elements.
- Section C will collect data on Gastrointestinal symptoms and gut health using Rome IV diagnostic questionnaire.
- Section D will collect data on diet and lifestyle factors

- **Data Analysis:**
  - The data collected will be tabulated using Microsoft excel and responses will be calculated as percentages.
  - Mean, Standard deviation and p value will be calculated using Microsoft excel and SPSS software v.25
  - p Value less than 0.05 will be considered statistically significant
  - Correlation between sleep pattern and gut health will be analyzed by Chi Square test using SPSS software v.25

**Research Topic:**

This cross-sectional study examines the association between sleep quality and gut disturbances. Since sleep is a crucial for cognitive performance and memory retention, poor sleep quality can impact academic success. Understanding the impact of sleep on gut brain axis can guide effective and timely interventions at the initial stages of the undergraduate courses and enhance their mental and physical wellbeing.

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